Part Number	63-5531-WH	**Document No.**	WI-0638
Part Name	Engine Bay Wiring Harness	**Revision**	A
Operation	Op 30: Harness Assembly and Electrical Test	**Effective Date**	2026-10-05
Work Center	Harness Board 7	**Takt / Cycle Time**	11.0 min
Prepared By	Manufacturing Engineering	**Approved By**	Quality Manager

1. Purpose and Scope

This work instruction defines the layout, taping, connector assembly, and 100 percent electrical test of wiring harness 63-5531-WH on the dedicated layout board at Harness Board 7.

2. Safety Requirements and PPE

Requirement	Detail
PPE	Safety glasses, finger cots or thin nitrile gloves for tape work.
Electrical test	The continuity tester is low voltage and safe to touch. Only the certified tester fixture T-6355 may be used.
Repetitive motion	Follow the 50 minute work / 10 minute rotation schedule posted at the cell.
Sharp tools	Use the guarded harness scissors only. No open blades at the board.

3. Tools, Equipment, and Materials

Item	Description	Spec / Calibration
Layout board	Board 63-5531, pin layout rev A	Board rev matches WI rev
Test fixture	Continuity and seating tester T-6355	Self-test pass at shift start
Tape dispenser	PVC harness tape, 19 mm	Tape lot in date
Pull test scale	Digital force gage, 0 to 200 N	Calibrated
Label printer	Thermal transfer, station printer	Ribbon OK, test label legible

4. Components / Materials Required

Component	Part / Spec No.	Qty
Lead set, cut and terminated	63-5531-LS (kitted)	1 kit
Connector, 12-way sealed	CN-1248-GY	2
Connector, 4-way sealed	CN-0427-BK	3
Convolute tubing, 10 mm	CT-1050	1.2 m
Edge clips	CL-0880	6
ID label	LB-6355 (printed at station)	1

5. Work Steps

Step	Instruction	Key Point / Specification
1	Verify the kit number on the lead set bag matches the work order, then empty the kit into the parts tray.	Kit 63-5531-LS, correct rev

Step	Instruction	Key Point / Specification
2	Route the main trunk leads along the board path, seating each lead in the board forks in the numbered order printed on the board.	All forks filled in order
3	Route branch A (12-way, left) and branch B (12-way, right) along their marked paths.	No crossed leads in the trunk
4	Insert terminals into the first 12-way connector per the pinout card C-1248 at the station. Push until a click is heard, then tug.	Click + tug test every cavity
5	Repeat terminal insertion for the second 12-way and all three 4-way connectors.	Click + tug test every cavity
6	Perform a pull test on two random terminals per connector using the force gage.	Min 40 N, no movement
7	Slide the convolute tubing over the main trunk between board marks T1 and T2.	Tubing covers T1 to T2 fully
8	Tape the trunk with half-lap wraps from T1 to T2 over the tubing ends, and spot tape each branch takeout.	Half-lap, no wire visible at ends
9	Install the six edge clips at the positions marked on the board.	Clip heads fully snapped
10	Remove the harness from the board and plug all connectors into test fixture T-6355.	All connectors fully latched
11	Run the automated test (continuity, shorts, and connector seating).	Fixture shows PASS (green)
12	On a FAIL, re-seat the flagged cavity once and re-test. A second FAIL is a nonconformance, tag and quarantine.	Max 1 re-test
13	Print and apply the ID label to the trunk 50 mm from the T1 mark. The label prints only after a PASS result.	Label placed at T1 + 50 mm
14	Coil the harness in three loops, secure with one tape wrap, and place in the outbound bin, max 15 per bin.	No kinks, 15 per bin max

6. Quality Checks

Characteristic	Specification	Method	Frequency
Electrical test	100% PASS on T-6355	Automated fixture	Every harness
Terminal retention	Min 40 N	Force gage, 2 per connector	Every harness
Tape coverage	Half-lap, T1 to T2	Visual vs board marks	Every harness
Label	Correct PN, legible barcode	Scan check at station	Every harness
First piece	All above + pinout audit	Full check + log	Start of shift

7. Nonconforming Material

If any check fails or a defect is observed, STOP the operation. Tag the part with a red NONCONFORMING tag, record the part number, lot, and defect on the tag, and move the part to the quarantine rack. Notify the line lead and Quality immediately. Do not rework or scrap without an approved disposition.

8. Revision History

Rev	Date	Description	Approved
A	2026-10-05	Initial release	QM

Operator Signature: _____ Date: _____

Supervisor Verification: _____ Date: _____